

RAG Pipeline

# Model Comparison Report

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## Pipeline Configuration

| Configuration               |                              |
|-----------------------------|------------------------------|
| Retrieval mode              | hybrid                       |
| Reranker                    | OFF                          |
| Adaptive weights            | ON                           |
| Fusion strategy             | RRF                          |
| BM25 / Sem weights          | 0.5 / 0.5 (ADAPTIVE)         |
| Filtering                   | OFF                          |
| Cross-modal boost           | ON                           |
| text_k / rerank_k / image_k | 10 / 5 / 5                   |
| Embed model                 | BAAI/bge-large-en-v1.5       |
| Image embed model           | ViT-L-14 (laion2b_s32b_b82k) |
| LLM model(s)                | gemma4:e2b                   |
| Chunk tokens / overlap      | 384 / 128                    |

## Metric Comparison — All Models

| ■■ RETRIEVAL (shared across all models) ■■            |             |
|-------------------------------------------------------|-------------|
| M1 Embedding time ↓                                   | 11.8317 s   |
| M2 Index size                                         | 176 vectors |
| M3 Retrieval latency ↓                                | 0.0460 s    |
| M4 Fused Relevance (Text)                             | 0.7489      |
| M4 Fused Relevance (Image)                            | 0.4175      |
| M5 Page Coverage@k ↑                                  | 92.86 %     |
| M9 Context length                                     | 3221 ch     |
| ■■ GENERATION QUALITY (per model, vs ground truth) ■■ |             |
| M6 ROUGE-1 ↑                                          | 0.5105      |
| M7 ROUGE-2 ↑                                          | 0.1938      |
| M8 ROUGE-L ↑                                          | 0.2904      |
| M10 BLEU ↑                                            | 0.1273      |
| M11 METEOR ↑                                          | 0.3523      |
| M12 BERTScore ↑                                       | 0.8769      |
| M13 FCD ↓                                             | 15.03       |
| M14 Faithfulness ↑                                    | 90.21 %     |
| M15 GT Coverage ↑                                     | 97.38 %     |
| ■■ PERFORMANCE (per model) ■■                         |             |
| M16 E2E Latency ↓                                     | 20.4969 s   |
| M17 Throughput ↑                                      | 0.049 q/s   |
| M18 CPU Usage ↓                                       | 0.24 %      |
| M19 RAM Usage ↓                                       | 0.127 GB    |
| GPU avg                                               | 28.40 %     |
| GPU peak                                              | 100.00 %    |

\* Retrieval metrics are computed once and are identical across model columns because retrieval runs before any LLM is invoked.